ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

ELECTROTECHNICS N6

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central N6 Electrotechnics N6 paper 1 · practice a.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct methods (e.g. per-phase calculations).

QUESTION 1

1.1 $E = \frac{Z\Phi Np}{60a}$, lap: $a = p = 4$ ✓; $E = \frac{720 \times 0{,}025 \times 1\,000 \times 4}{60 \times 4}$ ✓✓ $= 300$ V ✓ (4)

1.2.1 $I_f = 220 \div 110 = 2$ A ✓; $I_a = 32 - 2 = 30$ A ✓ (2)

1.2.2 $E_b = 220 - 30 \times 0{,}4$ ✓ $= 208$ V ✓ (2)

1.2.3 $T = \frac{E_b I_a}{2\pi N/60} = \frac{208 \times 30}{2\pi \times 1\,000/60}$ ✓✓ $= 59{,}59$ N·m ✓ (3)

1.2.4 $E_{b2} = 220 - 50 \times 0{,}4 = 200$ V ✓; $N_2 = 1\,000 \times \frac{200}{208}$ ✓ $= 961{,}54$ r/min ✓ (3)

1.3.1 Armature: $30^2 \times 0{,}4 = 360$ W ✓; field: $220 \times 2 = 440$ W ✓ (2)

1.3.2 Input $= 220 \times 32 = 7\,040$ W ✓; output $= 7\,040 - 360 - 440 - 500$ ✓ $= 5\,740$ W ✓; $\eta = \frac{5\,740}{7\,040} \times 100 = 81{,}53\%$ ✓ (4)

[20]

QUESTION 2

2.1.1 $V_{ph} = \frac{400}{\sqrt{3}} = 230{,}94$ V ✓; $Z = \sqrt{(12^2 + 16^2)} = 20\ \Omega$ ✓; $I_L = I_{ph} = \frac{230{,}94}{20}$ ✓ $= 11{,}55$ A ✓ (4)

2.1.2 $\cos\varphi = \frac{R}{Z} = \frac{12}{20}$ ✓ $= 0{,}6$ lagging ✓ (2)

2.1.3 $P = \sqrt{3} \times 400 \times 11{,}55 \times 0{,}6$ ✓ $= 4\,800$ W ✓; $Q = \sqrt{3} \times 400 \times 11{,}55 \times 0{,}8 = 6\,400$ var ✓; $S = 8\,000$ VA ✓ (4)

2.2.1 $\varphi_1 = \cos^{-1} 0{,}7 = 45{,}57°$; $Q_1 = 50 \tan 45{,}57°$ ✓ $= 51{,}01$ kvar ✓; $\varphi_2 = \cos^{-1} 0{,}95 = 18{,}19°$; $Q_2 = 50 \tan 18{,}19°$ ✓ $= 16{,}43$ kvar ✓; $Q_C = 51{,}01 - 16{,}43 = 34{,}58$ kvar ✓ (5)

2.2.2 Per phase: $34\,580 \div 3 = 11\,527$ var ✓; delta, so $V = 400$ V: $C = \frac{Q}{2\pi f V^2} = \frac{11\,527}{2\pi \times 50 \times 400^2}$ ✓ $= 229{,}29$ µF ✓ (3)

2.2.3 Any TWO: lower current, so smaller cables and switchgear ✓; lower kVA demand charges / electricity bill ✓; lower losses and voltage drop; more spare capacity on transformers (2)

[20]

QUESTION 3

3.1.1 $N_1 = 80 \times \frac{11\,000}{400}$ ✓ $= 2\,200$ turns ✓ (2)

3.1.2 $\Phi_m = \frac{E_2}{4{,}44fN_2} = \frac{400}{4{,}44 \times 50 \times 80}$ ✓✓ $= 0{,}0225$ Wb $(22{,}52$ mWb) ✓ (3)

3.2.1 Output $= 100 \times 0{,}8 = 80$ kW ✓; losses $= 1{,}2 + 1{,}8 = 3$ kW ✓; $\eta = \frac{80}{80+3} \times 100$ ✓ $= 96{,}39\%$ ✓ (4)

3.2.2 Output $= 0{,}5 \times 100 \times 1 = 50$ kW ✓; Cu loss $= 0{,}5^2 \times 1{,}8 = 0{,}45$ kW ✓; $\eta = \frac{50}{50 + 1{,}2 + 0{,}45} \times 100$ ✓ $= 96{,}81\%$ ✓ (4)

3.2.3 Maximum efficiency when Cu loss = iron loss ✓: $x = \sqrt{(\frac{1{,}2}{1{,}8})}$ ✓ $= 0{,}8165$ ✓; load $= 81{,}65$ kVA ✓ (4)

3.3 The losses (copper loss depends on current, iron loss on voltage) do not depend on the power factor ✓✓; the manufacturer does not know the power factor of the load, so the rating is given in volt-amperes ✓ (3)

[20]

QUESTION 4

4.1.1 $N_s = \frac{120 \times 50}{4}$ ✓ $= 1\,500$ r/min ✓ (2)

4.1.2 $N_r = 1\,500(1 - 0{,}04)$ ✓ $= 1\,440$ r/min ✓ (2)

4.1.3 $f_r = 0{,}04 \times 50$ ✓ $= 2$ Hz ✓ (2)

4.2.1 $15 - 0{,}8$ ✓ $= 14{,}2$ kW ✓ (2)

4.2.2 $0{,}04 \times 14{,}2$ ✓ $= 0{,}568$ kW $(568$ W) ✓ (2)

4.2.3 $P_{mech} = 14{,}2 - 0{,}568 = 13{,}632$ kW; output $= 13{,}632 - 0{,}3 = 13{,}332$ kW ✓✓; $\eta = \frac{13{,}332}{15} \times 100 = 88{,}88\%$ ✓ (3)

4.2.4 $T = \frac{13\,332}{2\pi \times 1\,440/60}$ ✓✓ $= 88{,}41$ N·m ✓ (3)

4.3 At starting, the stator windings are connected in star, so each winding receives $V_L/\sqrt{3}$ ✓; this reduces the phase current ✓; when the motor approaches full speed it is switched to delta for normal running ✓; the starting line current (and torque) is reduced to one-third of the direct-on-line value ✓ (4)

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QUESTION 5

5.1.1 $V_{ph} = \frac{11\,000}{\sqrt{3}} = 6\,350{,}85\text{ V}$ ✓; $I = \frac{2 \times 10^6}{\sqrt{3} \times 11\,000}$ ✓ $= 104{,}97\text{ A}$ ✓ (3)

5.1.2 $I = 104{,}97\angle -36{,}87°\text{ A}$ ✓; $E = V + jX_sI$ ✓; $jX_sI = 10 \times 104{,}97\angle 53{,}13° =$ $629{,}82 + j839{,}76\text{ V}$ ✓; $E = 6\,980{,}67 + j839{,}76$ ✓; $|E| = 7\,031{,}02\text{ V per phase}$ ✓ (5)

5.1.3 $\text{Regulation} = \frac{7\,031{,}02 - 6\,350{,}85}{6\,350{,}85} \times 100$ ✓ $= 10{,}71\%$ ✓ (2)

5.1.4 V as reference, I lagging V by 36,87° ✓; IX_s drawn at 90° leading I, with E as the phasor sum V + IX_s ✓ (2)

5.2 Any FOUR ✓✓✓✓: equal voltage (RMS); equal frequency; same phase sequence; voltages in phase at the moment of closing (synchronised, e.g. by synchroscope or lamps); same waveform (4)

5.3 Under-excitation: the motor draws lagging current ✓; increasing the excitation (over-excitation) makes the motor take a leading current ✓, so it supplies reactive power and improves the power factor of the whole system ✓; application: synchronous condenser for power factor correction in a factory ✓ (4)

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TOTAL: 100

END OF MARKING GUIDELINE